

Lecture Syllabus

Module Code	Semester	ECTS	SWS	Lecture	Tutorial	Lab
MECH-B-5-ANT-ANT-ILV	5	5	3	2	0	1
Course Name	Drive Technology					
Lecturer	D. T. McGuiness, Ph.D (Daniel.McGuiness@mci.edu) (4A-434c)					
Study Programme	Mechatronic Design Innovation					
Official Name	Antriebstechnik				Lingo	English
Lecture Prerequisites	The student should have a good background in circuit analysis and familiarity with integral/differential calculus. After completion, the student will have a good grasp of electrical machines from their construction, how they are designed, what types out there and what applications are possible.					
Course Objectives	The goal of this lecture is to give an introduction to the fascinating world of electric machines. We will start from understanding how they work to the types currently used in commercial and industrial applications. The lecture will focus on mathematical and engineering concepts which will have direct use with real-life engineering applications.					
Primary Course Content	Lecture Homepage on GitHub WebBook					
Secondary Course Content(s)	<i>Advanced Electric Drives: Analysis Control and Modeling Using MATLAB/Simulink</i> by Mohan Ned. , <i>Analysis of Electric Machinery and Drive Systems</i> by Krause Paul C. et. al. , <i>Design of Rotating Electrical Machines</i> by Pyrhonen Juha et. al , <i>Electric Machinery Fundamentals (5th Edition)</i> by Stephen J. Chapman. , <i>Electric Machinery</i> by Fitzgerald A. E. et. al. , <i>Electric Motors and Drives: Fundamentals Types and Applications</i> by Hughes A. et. al. ,					
Homework(s) and Project(s)	Quiz (5) Personal Assignment (15) Final Exam (30) Lab Work (50)					
Assessment Criteria	Assignment Type			Effect	Count	
	Quiz			5	1	
	Personal Assignment			15	1	
	Final Exam			30	1	
	Lab Work			50	4	

Lecture Structure

Order	Topic	Units	Self Study
1	Magnetic Circuits and Materials	4	4
	<i>Introduction • Flux Linkage, Inductance, and Energy • Magnetic Material Properties • AC Excitation</i>		
2	Transformers	4	4
	<i>The Stationary Electric Machine • Transformers under No-Load Conditions • The Effect of Secondary Current on an Ideal Transformer • Reactances and Types of Equivalent Circuits • Understanding Transformer Behaviour • Auto and Multiwinding Transformers • Poly-Phase Transformers • Voltage and Current Transformers</i>		
3	Electromechanical Energy Conversion (Self Study)	4	8
	<i>The Necessary Definitions • Forces and Torque under Magnetic Field • The Theory of Energy Balance • Energy in a Singly Excited Magnetic Circuit • Determining Force and Torque from Energy • Determination of Magnetic Force and Torque from CoEnergy • Multiply Excited Magnetic Systems</i>		
4	Lab 1: Installation of a Drive System	3	6
	<i>Hands-on Laboratory Work</i>		
5	Rotating Magnetic Fields (Self Study)	2	4
	<i>Fundamental Concepts • MMF of Distributed Windings • Magnetic Field in Rotating Machinery • Describing Rotating MMF Waves in AC Machines • Generated Voltage • Torque in Non-Salient-Pole Machines • Magnetic Saturation • Leakage Flux</i>		
6	Lab 2: Building a Lathe	4	8
	<i>Hands-on Laboratory Work</i>		
7	Synchronous Machines	4	8
	<i>To Spin with Synchronicity • Equivalent Circuits • Open and Short Circuit Characteristics • Steady-State Power-Angle Characteristics • Steady-State Operating Characteristics</i>		
8	Poly-phase Induction Drives	4	8
	<i>Introduction to Induction Machines • Current and Fluxes in Induction Machines • Induction Machine Equivalent Circuit • Analysis of the Equivalent Circuit • Calculating Power and Torque using Thevenin • Parameters Determination from No-Load and Blocked-Rotor Tests • The Effects of Rotor Resistance</i>		
9	Lab 3: Synchronous and Induction Drives	4	8
	<i>Hands-on Laboratory Work</i>		
10	DC Drives	4	8

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Order	Topic	Units	Self Study
	<i>Introduction · The Process of Commutation · Analytical Study · Armature Reaction · Steady-State Performance · Permanent Magnet Motors · Commutation and Interpoles · Compensation Windings · Series Universal Motor</i>		
11	Special Machines	2	3
	<i>Single-Phase Induction Motors · Shaded Pole · Brushless DC Motors · Switched Reluctance Motors</i>		
12	Lab 4: Photovoltaic Cells	4	8
	<i>Hands-on Laboratory Work</i>		
13	Final Exam	2	4
	<i>Written Exam</i>		
14	Total	45	90

- For two (2) sessions there will be a self-guided study which will require you to work through the material on your own.
- In these sessions you should focus on two topics: **Electromechanic Energy conversion** and **Rotating Magnetic Fields**. You can either access this content on WebBook or LectureBook which hold the **same** content.
- The next face-to-face session will focus on a recap of these topics along with any questions you may have.
- Afterwards there will be a 15 min long quiz covering these two topics **only**.

- Any major announcements will be made on SAKAI regarding any possible date/content/structural changes for the assignment(s), exam(s).
- Any lecture material will be posted at the lectures corresponding GitHub home-page. The link will be present on the lectures SAKAI homepage.
- If there are any questions regarding course content/exams/assignments please do not refrain from contacting me (Daniel.McGuiness@mci.edu).